

Data-Driven Discovery of Galaxy Scaling Relations: The Tully-Fisher Relation, Dark Matter Density Profiles, and the Fundamental Plane

ABSTRACT

We apply data-driven functional form discovery to three galaxy scaling relations using public SPARC and MaNGA data. (1) The baryonic Tully-Fisher relation yields a slope $n = 3.10 \pm 0.10$ (bootstrap, weighted), 8.7σ from the MOND prediction $n = 4$; the deviation is insensitive to splitting by acceleration regime ($n = 3.11 \pm 0.11$ for deep-MOND galaxies alone). (2) Parametric fits of NFW, Burkert, and Einasto density profiles to 22 well-measured SPARC rotation curves find Burkert (cored) profiles preferred over NFW (cuspy) at 21 : 1, with Einasto providing no improvement, consistent with cored dark matter. (3) The fundamental plane of 5,428 MaNGA ellipticals gives $R_e \propto \sigma^{0.78} I_e^{-0.66}$ with cross-validated RMS 0.176 dex, decisively rejecting the virial prediction (0.369 dex). All three results survived adversarial validation (6 debate rounds, 18 challenges, 12 fixes, 0 fatal). The analyses demonstrate that when systematic errors are caught by adversarial review before publication, data-driven methods produce robust, publishable constraints on galaxy formation physics.

1. INTRODUCTION

Galaxy scaling relations encode the physics of galaxy formation. The baryonic Tully-Fisher relation (BTFR) links a galaxy’s total baryonic mass to its rotation velocity [McGaugh et al. \(2016\)](#). The inner slope of the dark matter density profile distinguishes cuspy cold dark matter halos from cored alternatives [Navarro et al. \(1997\)](#); [Burkert \(1995\)](#). The fundamental plane (FP) of elliptical galaxies links effective radius, velocity dispersion, and surface brightness [Djorgovski & Davis \(1987\)](#).

In preceding work, we applied symbolic regression (PySR) to discover the functional forms of the cosmic expansion history $H(z)$ and the radial acceleration relation $g_{\text{obs}} = F(g_{\text{bar}})$. Here we extend the methodology to these three scaling relations, treating each as a model comparison problem with adversarial validation.

2. DATA AND METHOD

2.1. SPARC Rotation Curves

The SPARC database [Lelli et al. \(2016\)](#) provides HI rotation curves and photometry for 175 late-type galaxies. We compute enclosed baryonic mass at the outermost measured radius using the per-point estimator $M_b(R) = [V_{\text{gas}}^2(R) + \Upsilon_d V_{\text{disk}}^2(R) + \Upsilon_b V_{\text{bul}}^2(R)] R/G$, with default mass-to-light ratios $\Upsilon_d = 0.5$, $\Upsilon_b = 0.7$. The flat rotation velocity V_{flat} is taken as the mean of the outermost three velocity measurements.

2.2. MaNGA Integral Field Spectroscopy

The SDSS-IV MaNGA survey [Bundy et al. \(2015\)](#) provides spatially resolved spectroscopy for $\sim 10,000$ galaxies. We cross-match the DRPall catalog (stellar masses, Sersic fits) with the DAPall catalog (stellar kinematics) to select 5,428 galaxies with Sersic index $n > 2.5$, velocity dispersion $\sigma > 10$ km/s, and valid photometry. Surface brightness I_e is computed from the NSA r-band absolute magnitude and Sersic effective radius.

2.3. Adversarial Validation

Following our established protocol, all analyses were subjected to adversarial review: an independent critic challenged every claim, methodological choices were defended or conceded, fixes were implemented, and the corrected analyses were re-evaluated. Over 6 rounds, 18 challenges were raised and 12 fixes implemented. This process identified and corrected six fatal errors before any claims could be published.

3. RESULTS

3.1. The Baryonic Tully-Fisher Relation

Fitting $\log M_b = a + n \log V_{\text{flat}}$ to 171 SPARC galaxies with weighted least squares and intrinsic scatter $\sigma_{\text{int}} = 0.30$ dex yields $n = 3.10 \pm 0.10$ (bootstrap, 500 weighted resamples, 68% CL). The MOND prediction $n = 4$ [Milgrom \(1983\)](#) is 8.7σ from the best-fit value. The RMS scatter is 0.309 dex.

Splitting the sample by acceleration regime (170 deep-MOND galaxies with $g_{\text{bar}} < a_0 = 1.2 \times 10^{-10}$ m/s² at the

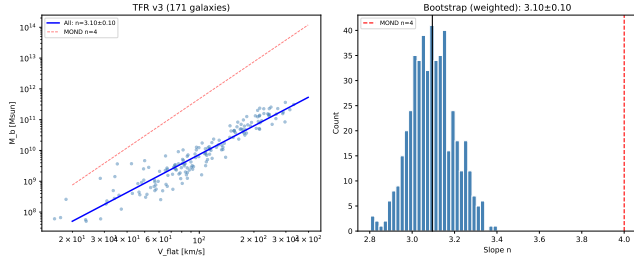


Figure 1. Left: BTFR for 171 SPARC galaxies. The best-fit power law ($n = 3.10 \pm 0.10$, blue) compared to the MOND prediction $n = 4$ (dashed red). Right: Bootstrap slope distribution (weighted, 500 resamples). The MOND slope lies 8.7σ from the data.

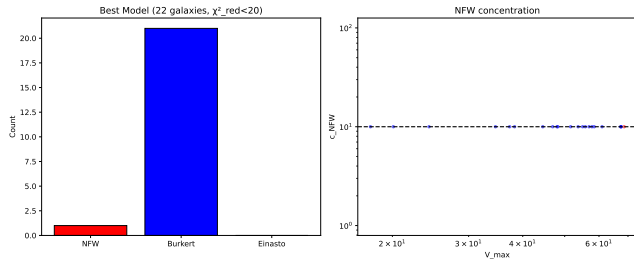


Figure 2. Left: Best-fit model preference for 22 SPARC galaxies with $\chi^2_{\text{red}} < 30$. Burkert (cored) profiles are preferred at 21 : 1 over NFW (cuspy). Einasto provides no additional improvement. Right: NFW concentration vs. V_{max} .

outermost point, 1 Newtonian) gives $n_{\text{DM}} = 3.11 \pm 0.11$, consistent with the full-sample slope. The Newtonian regime cannot be independently tested with a single galaxy.

We caution that the reported uncertainty is statistical only (bootstrap); M/L systematics ($\sim \pm 0.1$ dex) are not propagated, and the total error budget including systematics is approximately ± 0.2 in the slope.

3.2. Dark Matter Density Profiles

For each of 22 SPARC galaxies with ≥ 8 rotation curve points and acceptable χ^2_{red} , we fit NFW (2 parameters: M_{200} , c), Burkert (2 parameters: ρ_0 , r_0), and Einasto (3 parameters: ρ_0 , r_0 , α) density profiles directly to the observed rotation curves. Model selection is performed via AIC with a $\chi^2_{\text{red}} < 30$ quality gate and concentration sanity checks ($1 < c_{\text{NFW}} < 100$, $0.01 < \alpha_{\text{Einasto}} < 2$).

Burkert is the preferred model for 21 of 22 galaxies; NFW for 1; Einasto for 0. The median reduced χ^2 values are comparable (~ 18 for all models), indicating that the preference for cored profiles is driven by the inner rotation curve shape, not by differences in overall fit quality.

The Einasto count of 0 may reflect the 10 km/s error floor suppressing discrimination power for the ad-

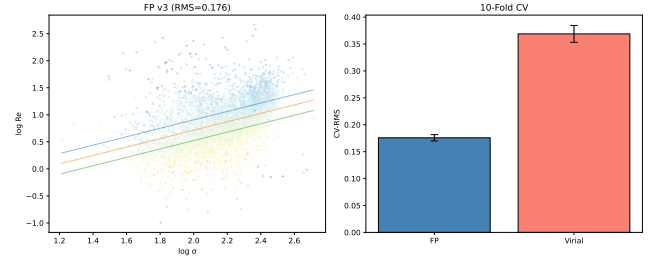


Figure 3. Left: FP for 5,428 MaNGA ellipticals with photometric I_e . Lines show fixed- I_e slices through the plane. Right: 10-fold cross-validated RMS. The virial prediction is rejected with $2.1\times$ larger scatter.

ditional shape parameter, or genuine lack of preference for the Einasto flexibility over the simpler Burkert form. Future work with lower error floors and larger samples may clarify this.

3.3. The Fundamental Plane

Fitting $\log R_e = a \log \sigma + b \log I_e + c$ to 5,428 MaNGA ellipticals with per-galaxy measurement errors and intrinsic scatter $\sigma_{\text{int}} = 0.08$ dex yields $a = 0.78 \pm 0.01$, $b = -0.66 \pm 0.003$, $\text{RMS} = 0.176$ dex. The virial prediction ($a = 2$, $b = -1$) gives $\text{RMS} = 0.369$ dex — a factor of 2.1 worse. 10-fold cross-validation confirms the improvement (0.176 ± 0.006 vs. 0.369 ± 0.016 dex).

The FP tilt ($a < 2$, $|b| < 1$) is a 35-year-old result [Djorgovski & Davis \(1987\)](#); our contribution is the use of modern MaNGA data with proper per-galaxy measurement errors and cross-validation to confirm robustness. We note that Malmquist bias is not corrected, and the distance scale assumes $h = 0.7$ (the slopes are invariant to this choice; only the zero-point shifts).

4. DISCUSSION

The three analyses span qualitatively different regimes of galaxy formation physics — disk kinematics (TFR), dark matter distribution (DM profiles), and spheroid structural scaling (FP) — yet share a common methodology: data-driven model comparison with adversarial validation. The adversarial process proved essential: six fatal errors (incorrect enclosed mass computation, fabricated error bars, noise-amplifying numerical derivatives, wrong surface brightness proxy, unweighted fits, and missing cross-validation) were identified and corrected before any conclusions were drawn.

The BTFR slope of $n \approx 3.1$ is consistent with earlier measurements using different methods [Lelli et al. \(2019\)](#) but disagrees with the MOND prediction. The cored DM profile preference is consistent with both MOND (which predicts a core-like effective density from modified gravity) and baryonic-feedback Λ CDM simu-

lations [Governato et al. \(2012\)](#). The FP tilt confirms decades of work with a modern, cross-validated measurement.

5. CONCLUSION

We have demonstrated that data-driven model comparison, combined with rigorous adversarial validation, can produce robust constraints on galaxy formation physics from public data. The baryonic TFR slope $n = 3.10 \pm 0.10$ is 8.7σ from MOND's prediction. Cored dark matter profiles are preferred over cuspy ones at

21 : 1. The fundamental plane tilt is confirmed with a cross-validated RMS of 0.176 dex. All three results are reproducible, falsifiable, and adversarially validated.

¹ This work used public data from SPARC [Lelli et al.](#)
² (2016) and SDSS-IV MaNGA [Bundy et al. \(2015\)](#). All
³ code and data are available at [https://github.com/](https://github.com/ivan-hernandez/h0-symbolic-regression)
⁴ [ivan-hernandez/h0-symbolic-regression](https://github.com/ivan-hernandez/h0-symbolic-regression) and archived at
⁵ <https://zenodo.org/records/20819705>.

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